

FingerViP: Learning Real-World Dexterous Manipulation with Fingertip Visual Perception

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*Equal contribution

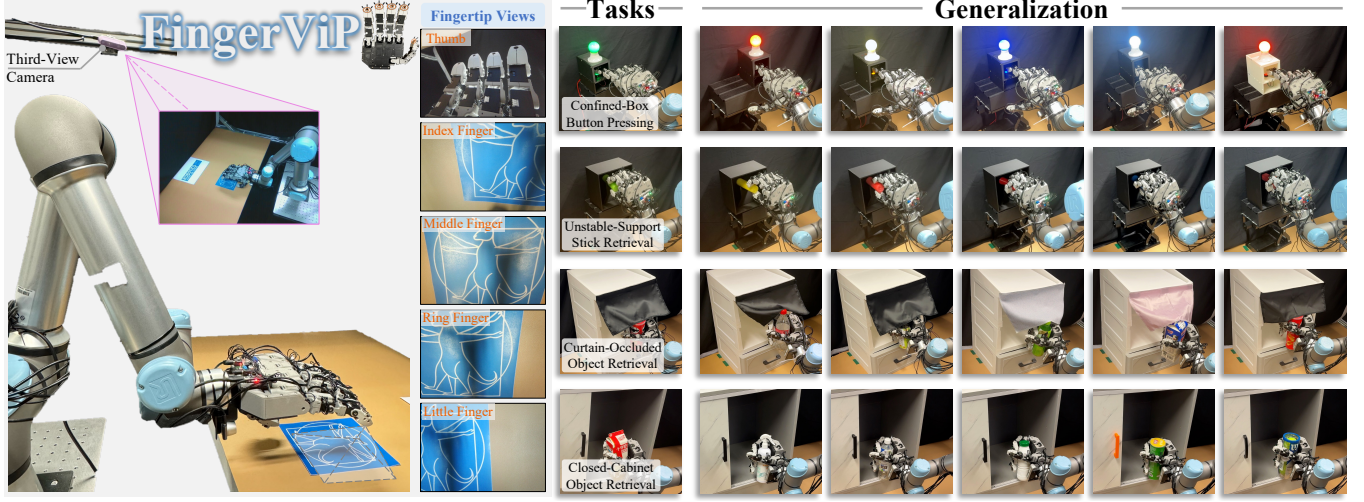


Fig. 1: We present **FingerViP**, a learning system that utilizes a visuomotor policy with fingertip visual perception to improve dexterous manipulation, especially under confined and highly-occluded settings. *Left*: We designed a vision-enhanced fingertip module and integrated it with a robotic hand, enabling comprehensive, hand-centric multi-view observations of both the hand and the surrounding environment. *Right*: Leveraging fingertip vision, our system achieves strong performance across diverse real-world tasks and scenarios, demonstrating high efficiency and robust generalization.

Abstract—The current practice of dexterous manipulation generally relies on a single wrist-mounted view, which is often occluded and limits performance on tasks requiring multi-view perception. In this work, we present FingerViP, a learning system that utilizes a visuomotor policy with fingertip visual perception for dexterous manipulation. Specifically, we design a vision-enhanced fingertip module with an embedded miniature camera and install the modules on each finger of a multi-fingered hand. The fingertip cameras substantially improve visual perception by providing comprehensive, multi-view feedback of both the hand and its surrounding environment. Building on the integrated fingertip modules, we develop a diffusion-based whole-body visuomotor policy conditioned on a third-view camera and multi-view fingertip vision, which effectively learns complex manipulation skills directly from human demonstrations. To improve view-proprioception alignment and contact awareness, each fingertip visual feature is augmented with its corresponding camera pose encoding and per-finger joint-current encoding. We validate the effectiveness of the multi-view fingertip vision and demonstrate the robustness and adaptability of FingerViP on various challenging real-world tasks, including pressing buttons inside a confined box, retrieving sticks from an unstable support, retrieving objects behind an occluding curtain, and performing long-horizon cabinet opening and object retrieval, achieving an overall success rate of 80.8%. All hardware designs and code will be fully open-sourced.

I. INTRODUCTION

Dexterous manipulation, ranging from daily-life household [42, 83] to industrial assembly [58, 65, 40], remains one of the most challenging yet fundamental problems in robotics. Early systems largely rely on simple two-finger parallel grippers, which limit fine-grained skills such as in-hand manipulation and tool use [43]. The increasing availability of commercial and open-source multi-fingered hands [2, 1, 57, 62] is lowering the hardware barrier and broadening the scope of dexterous manipulation research. Nevertheless, achieving human-level precision and robustness often requires more than advanced mechanics. Robust perception and intelligent control are essential for operating in complex, unstructured real-world environments, especially under severe occlusion and spatial constraints [5].

Recent advances in visuomotor policy learning for dexterous manipulation [63, 70, 13] have highlighted the benefits of vision-based perception for improving environmental awareness and action generation. Many existing systems use either external cameras [21, 12, 35] or a wrist-mounted camera [62, 31, 14], while others [73, 46, 11] combine both modalities. Despite their utility, external and wrist-mounted cameras can be unreliable in narrow, cluttered, and heavily occluded spaces. Specifically,

external cameras are distant and frequently occluded by the hand and objects, obscuring contact-level interactions [36, 20]. In contrast, wrist cameras are closer yet limited by a single viewpoint and are often occluded by the hand and the manipulated object during interaction (e.g., grasping), hindering localization and contact monitoring [67, 47, 80]. Moreover, the physical bulk of a wrist camera reduces clearance and restricts hand motion, which is particularly problematic in tight, confined settings. These limitations motivate a perception mechanism that provides close-range, multi-view observations without adding additional wrist-mounted hardware.

To address this challenge, we design a vision-enhanced fingertip module that embeds a miniature camera in each fingertip of a dexterous hand. Compared with external and wrist-mounted cameras, fingertip vision provides a close-range view near contact, capturing fine-grained interaction cues (e.g., local texture) that are often occluded from distal viewpoints. This contact-proximal view also supports reliable estimation of contact states (e.g., contact onset and slip) during interaction, whereas vision-based tactile sensors typically require costly specialized hardware and remain limited to the contact patch [38, 71]. Equipping multiple fingertip modules further enables multi-view perception, yielding more comprehensive observations of both the hand and the surrounding scene. Moreover, the fingertip module preserves the hand’s form factor and avoids interference in tight-clearance manipulation, while providing rich semantic cues in confined and heavily occluded spaces.

Building on our fingertip design, we propose a learning-based framework to train a visuomotor policy directly from human demonstrations, enabling a systematic evaluation of the advantages of fingertip visual perception for dexterous manipulation. To address the challenges posed by dynamic, high-dimensional multi-fingertip inputs with continuously changing viewpoints, we introduce a transformer-based diffusion model [15] in which cross-attention captures semantic relationships between multi-fingertip image tokens and action tokens. Additionally, to preserve the spatial information and contact-related signals of fingertip vision, fingertip-view poses and per-finger joint currents are encoded and concatenated to visual features. By fusing fingertip (hand-centric) observations with third-view (global) context, the policy leverages complementary cues for both contact-rich interaction and scene-level awareness, facilitating robust policy learning from human demonstrations.

In summary, our main contributions are as follows:

- We develop a vision-enhanced fingertip module with an embedded miniature camera and deploy it on each finger of a multi-fingered hand, enabling hand-centric multi-view observations that effectively mitigate occlusions without significantly increasing the hand’s physical envelope.
- We propose FingerViP, a learning system that leverages third-view, and multi-view fingertip observations with fingertip-camera pose and per-finger joint-current encodings, to improve view–proprioception alignment and contact awareness, enabling robust dexterous manipulation.
- We demonstrate the effectiveness and generalization of

FingerViP through a range of challenging real-world tasks and scenarios.

II. RELATED WORKS

A. Perception for Multi-fingered Manipulation

Multi-fingered robotic systems utilize a range of sensors, with vision and tactile feedback being the most prominent, to perform complex and dexterous manipulation. In recent years, tactile sensors, mimicking the human skin, have gained significant attention, enabling contact-rich manipulation [22, 65, 32, 41]. Such systems have successfully executed tasks like thin-object picking, tight-clearance object insertion, and wiping. However, tactile sensors are costly and can only provide feedback upon making contact with an object [45, 76], meaning they are unable to detect the object’s position or motion beforehand. This limitation becomes particularly pronounced for long-horizon manipulation tasks that involve rich pre-contact phases such as approaching. Recent advances in visuomotor policy have shown promising performance in tackling complex tasks in an end-to-end way using visual observations [15, 75]. Multi-fingered hands are equipped with wrist-mounted camera [62, 19] and third-view cameras [3, 64, 21] for dexterous manipulation. By leveraging rich visual feedback from cameras, these approaches enable robots to perceive multi-finger joint states, object geometry, spatial relationships, and task-relevant context during long-horizon dexterous manipulation.

B. Human Demonstration Collection

For dexterous manipulation tasks, human demonstrations remain a widely adopted strategy for collecting task-relevant data [4, 6]. Previous works, such as GRAB [59] and ARCTIC [18], use marker-based MoCap system to collect human demonstrations, which are precise but expensive. With advances in vision-based hand and object pose estimation, recent works [53, 12, 29, 23, 77] have explored learning dexterous manipulation skills directly from human videos and transferring the learned skills to a real robot. However, such approaches tend to perform poorly in highly cluttered and severely occluded scenarios, where reliable hand pose estimation becomes difficult [82, 60, 26]. Moreover, due to the human–robot embodiment gap, accurate and robust skill transfer remains a significant challenge. To mitigate these issues, low-cost wearable gloves such as Dexumi [67], DEXOP [19], and Doglove [74] have been proposed for teleoperation and dexterous manipulation data collection in prior studies. Wang et al. [63] and Yang et al. [70] developed scalable and portable data collection systems for dexterous manipulation. While effective, their non-negligible weight and mechanical constraints can lead to user fatigue during prolonged operation.

More recently, vision-based teleoperation offers the possibility to endow robots with human-level intelligence to physically interact with the environment. Anyteleop [54] and Dexpilot [25] use low-cost camera sensors to solve manipulation tasks. In addition, with the rapid development and widespread adoption of commercial augmented reality (AR) devices, several works [69, 16, 34, 13] have leveraged AR headsets (e.g.,

Meta Quest3 and Apple Vision Pro) to directly capture hand poses and remap the estimated joint angles to robotic hands for teleoperation. This paradigm enables stable and accurate motion capture without physically constraining the human hand, offering improved user experience combined with AR interfaces, higher data collection efficiency, and more intuitive human–robot interaction.

C. Imitation Learning from Demonstrations

Imitation learning (IL) enables robots to mimic human behaviors through learning from experts. One of the simplest IL algorithms, Behavioral Cloning (BC), treats the problem of learning behavior as a supervised learning task [52]. The modeling methods commonly used in traditional BC, such as regression-based method (e.g., MSE [7]), discretization-based method [56, 39], and cluster-based methods (e.g., K-means [24, 8]), have limitations when modeling complex action distributions and struggle to effectively capture the diversity and nuances of human behavior [51]. In addition, some alternative IL paradigms have been proposed, including IRL [81], GAIL [27], and DAgger [56]. However, these IL paradigms typically involve high interaction costs, computational complexity, and limited scalability in real-world robotic systems [33, 48].

Over the past few years, Denoising Diffusion Probabilistic Models (DDPMs) [28] have emerged as a powerful generative framework and have been increasingly adopted in robotics for behavior modeling and control. Recent works [37, 51, 10, 49] extend diffusion models to motion planning [10], trajectory optimization [49], and offline reinforcement learning [37, 51], enabling multi-solution planning, long-horizon temporal consistency. Chi et al. [15] further extend diffusion models to visuomotor policy learning by directly modeling conditional action distributions, enabling robust learning of complex, multi-step manipulation policies from limited demonstrations under perceptual uncertainty. More advancements [79, 44, 72, 30] continuously explore the boundaries of possibilities for robot behavior generation. Recently, diffusion-based policies have been applied to dexterous manipulation with multi-fingered hands [64, 21], demonstrating promising performance in real-world tasks. Overall, these results highlight the strong potential of diffusion models for scalable and robust robot behavior generation.

III. METHODS

A. Robot Dexterous Hand

The robot dexterous hand setup in this work, as shown in Fig. 2 (a), is inspired by the open-source, low-cost RAPID Hand [62]. The hand is 3D-printed and servo-actuated, adopting an anthropomorphic architecture with five fingers and 20 DoFs (four per finger). Differential bevel gears at the metacarpophalangeal (MCP) and carpometacarpal (CMC) joints enable coupled flexion–extension and abduction–adduction, approximating the ball-and-socket motion. The thumb includes an additional CMC joint for opposition. Building on this platform, we developed a vision-enhanced fingertip module.

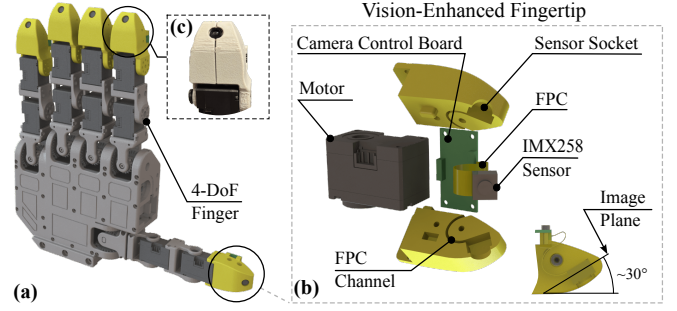


Fig. 2: **CAD Rendering.** (a) The robot dexterous hand. (b) Exploded view of the vision-enhanced fingertip structure. (c) A fabricated fingertip module.

B. Vision-Enhanced Fingertip Module

The vision-enhanced fingertip, which embeds a miniature camera as shown in Fig. 2 (b), is low-cost, compact, and highly modular, enabling quick fabrication, easy integration, and high maintainability.

Fingertip Camera Selection: Three key criteria were considered to select the fingertip cameras: (i) physical size, (ii) field of view (FOV), and (iii) connectivity. Given the limited fingertip volume, we choose a low-cost Generic 12MP SONY IMX258 USB Camera Module, which features a split sensor–PCB design connected by a short flexible printed circuit (FPC). Its compact sensor package ($8.5 \times 8.5 \times 4.95$ mm) enables easy integration within the fingertip. The controller PCB dimension ($30 \times 15 \times 1.6$ mm) sits within the fingertip envelope and can be mounted externally, making it a modular component. The camera provides a wide 135° FOV and supports autofocus. Moreover, its USB Video Class (UVC) compliant nature allows for plug-and-play over USB with no additional drivers.

Structural Modification and Camera Integration: The sensor is integrated into the fingertip within a pocket tilted by $\sim 30^\circ$ relative to the palmar surface so that the camera can jointly observe the region ahead of and beneath the finger. The fragile FPC-sensor junction is embedded inside the fingertip body, and the PCB is mounted externally on the dorsal side (fingernail position), improving the reliability of the FPC-to-PCB connection while keeping the fingertip compact and modular. Concretely, an internal S-shaped channel is kept to route the FPC. This channel guides the FPC to the PCB, which is mounted on 4 mm standoffs recessed by 3 mm to preserve the finger thickness. During assembly, the sensor and FPC are fixed in one half-shell and fully encapsulated when the second half is closed, assisted by the mortise–tenon alignment. The fabricated and assembled module is shown in Fig. 2 (c).

C. Teleoperation and Data Collection

Teleoperation: To efficiently collect human demonstrations, a high-DoF teleoperation interface [50] is employed to capture hand poses and enables real-time control of the robotic arm and hand. Specifically, an Apple Vision Pro tracks the operator’s hand at a frequency of 60 Hz and streams 21 hand keypoints and a 7-D wrist pose (i.e., 3D position and

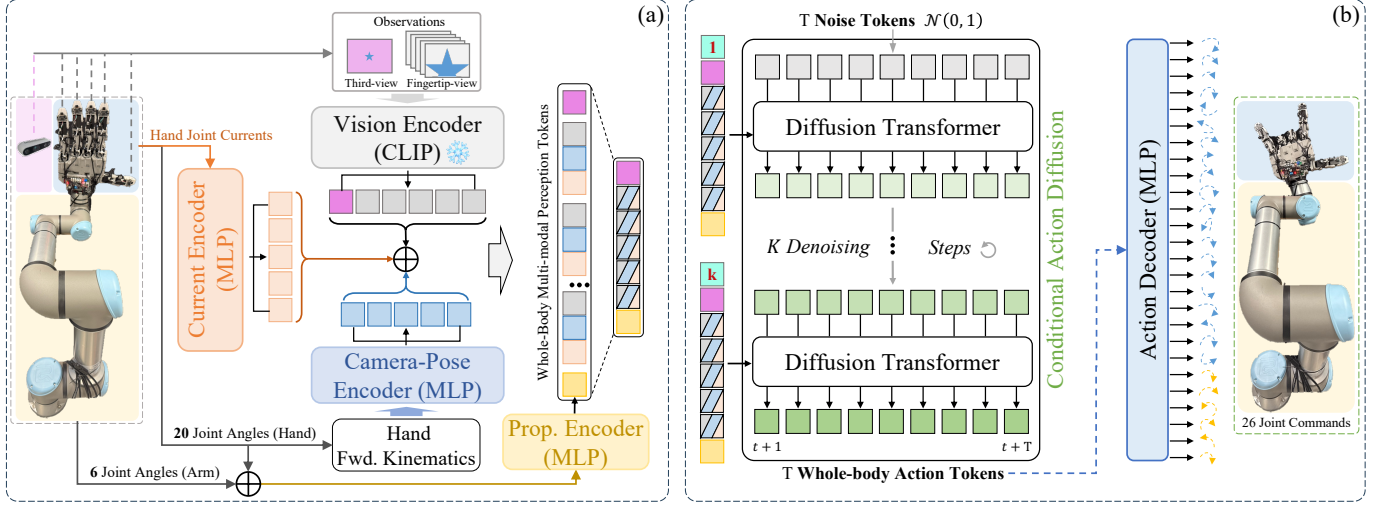


Fig. 3: **Whole-Body Visuomotor Policy with Fingertip Visual Perception.** (a) At time t , the policy observes five fingertip RGB images (gray) and one third-view image (pink), 20 hand joint currents, and 26 arm-hand joint angles. Fingertip camera poses are computed from hand forward kinematics. Visual features are extracted using a frozen vision encoder (CLIP), while camera poses, per-finger currents, and joint angles are embedded using MLP encoders and concatenated to construct the whole-body multi-modal perception tokens. (b) Conditioned on these tokens, a transformer-based diffusion model performs K denoising steps to generate a horizon of T whole-body action tokens, which are decoded by an MLP into 26-DoF joint commands.

quaternion orientation) over UDP protocol. To align each human keypoint with its corresponding robot hand link, a per-phalangeal segment geometric calibration is applied to reduce local kinematic discrepancies and mitigate global-scale distortions. The human hand motions are retargeted by solving an optimization problem with an objective consisting of a conformal-aligned constraint, a contact-aware coupling constraint, and temporal smoothness [62]. The joint angles of a UR5e arm are computed through inverse kinematics based on the wrist pose, enabling coordinated arm–hand control. The control system runs at 100 Hz and publishes 26 joint commands to the arm and hand.

Data Collection: In this work, we collect (i) target joint commands, (ii) robot proprioception: joint positions, and currents for all 26 DoFs (6 for arm and 20 for hand), and (iii) multi-view RGB observations from six cameras: five fingertip cameras and one third-view RGB-D camera (RealSense D435i). These multi-modal signals run at different frequencies. Specifically, all cameras operate at 30 Hz with a resolution of 640×480 , and all target joint commands are received at 100 Hz. However, precise temporal synchronization of multi-modal data is critical for collecting data and training policies from demonstration data for dexterous manipulation tasks. To address this, each fingertip camera is captured in an independent thread and synchronized via a global threading event, ensuring that all cameras are initialized before acquisition and that frames are temporally aligned across views. Each fingertip camera maintains a frame-queue buffer for parallel capture with low-latency access to the latest observation (6.7 ms average latency across five fingertip cameras). In addition, three dedicated threads are used to collect robot proprioception, fingertip images, and third-view images, respectively. Once control commands are dispatched by the main controller, proprioceptive data and all RGB streams are

recorded simultaneously under the same strategy. Finally, the resulting data are compressed and saved in Zarr format for post-processing.

D. Dexterous Visuomotor Policy

To control a 26-DoF robotic system, a transformer-based visuomotor policy is introduced and trained on human demonstrations to infer six joint angles of the arm and twenty joint angles of the hand simultaneously from six visual observations and proprioception. An overview (Fig. 3) of the policy is first introduced, followed by detailed explanations of the concrete modules.

Policy Overview: Our policy is designed based on the Diffusion Policy [15], which infers joint-space actions via denoising diffusion conditioned on the latest two steps of observation data. Concretely, at each time step t , the policy takes both the previous observation O_{t-1} and current observations O_t as conditional inputs and predicts an n -step action sequence $A_t = \{a_t, a_{t+1}, \dots, a_{t+n}\}$, where only $m \leq n$ actions are executed. In our implementation, we set $n = 16$ and $m = 8$. The observation at time step t is $O_t = \{I_t, P_t, C_t, Q_t\}$, where $I_t \in \mathbb{R}^{6 \times 224 \times 224 \times 3}$ consists of six RGB images from one third-view camera and five fingertip cameras, $P_t \in \mathbb{R}^{5 \times (3+6)}$ are the camera poses, $C_t \in \mathbb{R}^{20 \times 1}$ are whole-hand joint currents and $Q_t \in \mathbb{R}^{26 \times 1}$ are joint angles of both arm and hand. The actions are taken from human teleoperation commands during training and each a_t is represented as 26 joint angles.

Coordinating Fingertip Views With Positional and Joint Current Encoding. The robot’s motion continuously changes each fingertip camera’s pose and, consequently, its observing region, yielding a much larger observation space than static views (e.g., a fixed top-down camera). To better coordinate these moving views and improve policy learning efficiency, we introduce view-dependent positional encoding for each

fingertip camera. Specifically, we use Pinocchio to parse the hand URDF, identify individual links of the robot hand, and compute each fingertip camera pose in the hand base frame via the forward kinematics model $\mathcal{F}$ from the current hand joint angles $Q_t^h \subset Q_t$. Let ${}^B\mathbf{T}_t \in SE(3)$ denote the transformation from the hand base frame B to camera frame c at time step t :

$${}^B\mathbf{T}_{c,t} = [{}^B\mathbf{R}_{c,t} \ {}^B\mathbf{p}_{c,t}; \mathbf{0}^\top \ 1] = \mathcal{F}(Q_t^h), \quad (1)$$

where ${}^B\mathbf{p}_{c,t} \in \mathbb{R}^3$ is the camera position and ${}^B\mathbf{R}_{c,t} \in SO(3)$ the orientation. Following the rotation representation of Zhou et al. [78], we parameterize orientation by the first two columns of the rotation matrix,

$${}^B\mathbf{r}_{c,t} = \text{vec}({}^B\mathbf{R}_{c,t}[:, :2]) \in \mathbb{R}^6, \quad (2)$$

so that each camera pose is parameterized by a 3-D position vector and a 6-D rotation vector and represented by $({}^B\mathbf{p}_{c,t}, {}^B\mathbf{r}_{c,t}) \in \mathbb{R}^9$. The camera position and orientation are independently projected to 128-D features using linear layers and concatenated into a 256-D pose embedding. Beyond vision information, finger joint currents provide complementary signals that are tightly coupled to manipulation phases. During contact such as grasping, holding, or pressing, joint currents exhibit characteristic changes that reflect force loading, micro-slip, and local deformation, yielding distinctive contact cues that are difficult or impossible to acquire from RGB images alone under pre-contact uncertainty or severe occlusion. Therefore, per-finger joint currents are included as contact-aware features. Rather than encoding the whole-hand current vector, we project each finger's 4-D current vector independently to a 128-D feature via a linear layer to preserve finger-wise independence.

Transformer-based Multi-view Semantic Correspondence Learning. To efficiently extract task-relevant cues from five dynamically moving fingertip cameras and a fixed third-view camera, a pretrained vision encoder and a transformer decoder are used to extract and aggregate semantics across multiple views. Prior works [68, 14, 66] have shown that pretrained vision foundation models such as CLIP [55] and DINO [9] provide strong semantic representations and improve performance in visually complex robotic manipulation tasks. Therefore, a CLIP-pretrained model is deployed as the image encoder in this policy. Specifically, the raw 640×480 images are center-cropped to 224×224 and processed by a frozen CLIP ViT-B/16 encoder [17], yielding a 768-D class token feature per view. Features from the five fingertip cameras are linearly projected to 384-D using a shared linear projector, while the third-view feature is projected with a separate (view-specific) linear layer. Each projected 384-D visual feature is then concatenated with its corresponding camera-pose embedding and finger-current embedding to form a 768-D token per camera. Robot joint angles are embedded to 768-D and appended, producing a $(6 + 26) \times 768$ token sequence at time t . For action generation, we adopt a time-series diffusion transformer architecture [15] to iteratively denoise noisy action samples A_t^k , where k denotes the denoising step. At denoising step k , noisy action samples A_t^k are fed as input tokens, while

the observation tokens condition the model through multi-head cross-attention [61]. This design efficiently fuses multi-view vision and proprioception and learns the correspondence between observations and joint-space actions.

IV. EXPERIMENTS

To evaluate the capability of the proposed FingerViP, we conduct a set of challenging real-world dexterous manipulation tasks. To ensure fair comparisons, all methods for each task are trained using the same set of demonstrations and evaluated under identical initial robot states and scene configurations.

A. Experiment Setup

1) *Hardware*: We equipped a UR5e arm with the 20-DoF hand, and use an Intel RealSense D435i as the static third-view camera. For fair comparison, we additionally mounted a D435i on the wrist to record synchronized wrist-view observations during teleoperation. An Apple Vision Pro was employed to collect teleoperation commands. System control, data collection, and policy evaluation ran on a desktop PC (Intel i7-12700K CPU, RTX 4080 GPU), which was equipped with a PCIe USB expansion card providing an additional 20 Gbps of dedicated bandwidth to support reliable streaming of seven cameras.

2) *Comparisons*: The primary focus of this work is to underscore the ability of the proposed system with fingertip visual perception. To this end, we compare our policy against the following conditions across all tasks:

- *Wrist Camera (WC)*: Observations from a single wrist-mounted camera.
- *Third-View Camera (TVC)*: Observations from a single third-view camera.
- *Fingertip Cameras Only (FTC)*: Observations from five fingertip cameras.
- *TVC w/ WC*: Observations from both third-view camera and wrist-mounted camera.
- *FTC w/ WC*: Observations from five fingertip cameras and a wrist-mounted camera.
- *Human Teleoperation (HT)*: Robot teleoperated by a human using Apple Vision Pro.

B. Tasks

We validate FingerViP in real-world settings across four challenging dexterous manipulation tasks. Details of each task are provided in the following sections.

1) *Confined-Box Button Pressing*: The task requires the robot to press a button inside a box, where the space allows only a single finger to act. The box with a colored illuminated button at the bottom is open on one side, and a bulb with the same color on the top serves as the success indicator, as shown in Fig. 4 (a). The robot needs to approach and align with the box, fold the other four fingers, insert the index finger into the box, and then press the button to turn on the bulb. Task success is indicated only by whether the bulb is lit.

Challenges: Highly-constrained space: The confined space inside the box requires fine alignment and allows only one finger to reach the button and execute the press action.



Fig. 4: **Scenarios for the Four Challenging Real-World Tasks.** Each row lists the various cases for both training and testing.

Visual occlusions: The wrist-mounted and third-view cameras are nearly occluded when the other four fingers curl up and the extended finger enters the box. Once entered, the target button can only be observed and located by the index fingertip camera. **High precision and contact sensitivity:** Task success requires (i) precise approach and alignment between the index finger and the small box opening, and (ii) contact-aware control to execute the press, avoiding slipping or missing the button.

Test scenarios: We conduct **42** rollouts for each policy. As shown in Fig. 4 (a), the test scenarios can be grouped into the following six categories: *T1*) Variations in the initial button-box position (18 rollouts); *T2*) Buttons in unseen colors (6 rollouts); *T3*) Non-illuminated buttons (6 rollouts); *T4*) Buttons in unseen shapes (6 rollouts); *T5*) A white button box (3 rollouts); and *T6*) Changes in ambient light intensity (3 rollouts).

Performance: The training dataset contains 255 demonstration episodes, with an average duration of 7.2 s. We report both qualitative (Fig. 5 (a)) and quantitative results (Table I), our method achieves an overall 73.8% success rate, outperforming all baselines.

The WC policy performs the worst (Table I), due to the heavily occluded wrist view when the other four fingers are curled, making it difficult to complete the task. The TVC policy often fails to localize the box opening and internal button precisely under occlusion, leading to collisions with the box during insertion. The TVC w/ WC baseline, despite leveraging two views, still suffers from the occlusion problem. Compared to ours, the FTC baseline performs worse during the approach phase with large arm-hand motions, which is consistent with the limited global scene awareness without a true global view (i.e., a third-view camera). The FTC w/ WC baseline degrades further compared to FTC. In this task, the insertion-relevant local cues come primarily from the index-finger view, as the other fingers are fully curled and their fingertip views are largely self-occluded. The wrist view is also blocked by the curled fingers, providing limited useful information.

2) *Unstable-Support Stick Retrieval:* The task requires the hand to use three fingers, the thumb, index, and middle finger, to grasp a foam craft stick and retrieve it from its unstable support. The lightweight stick is loosely supported by a slender U-shaped cradle inside a box, as shown in Fig. 4 (b). Task success is defined as fully pulling the stick out from the support

without dropping it.

Challenges: Constrained space: The robot can only place three extended fingers inside the box to avoid collision with the wall. Unstable support: The robot needs to contact and grasp the foam stick carefully without knocking it off the slender cradle. Multi-contact coordination: Successful retrieval requires coordinated, stable contacts across multiple fingers during grasping and pulling; insufficient or unstable contact can cause the stick to slip or drop. Precision requirements: The three fingers must maintain an open shape configured to center the stick, then accurately close the fingertips to conform to the stick surface for a secure grasp.

Test scenarios: We conduct **37** rollouts for each policy. As shown in Fig. 4 (b), the test configurations can be grouped into five categories: *T1*) Variations in the initial box positions (18 rollouts); *T2*) Foam sticks in unseen colors (6 rollouts); *T3*) Foam sticks in unseen shapes (6 rollouts); *T4*) Varied stick material (thermoplastic polyurethane, TPU; 3 rollouts); and *T5*) Foam sticks with unseen lengths (4 rollouts).

Performance: The training dataset contains 256 demonstration episodes, with each demonstration averaging 6.1 s. We report both qualitative (Fig. 5 (b)) and quantitative results (Table I), our method achieves an overall 75.7% success rate, outperforming all baselines.

The TVC baseline can roughly localize the stick, while struggles to keep it centered during the approach and pre-contact phases, resulting in frequent knocking off the stick due to the absence of reliable local visual feedback. The WC policy often fails to localize the stick precisely and to maintain stable contact, leading to frequent misses or knock-offs. Although TVC w/ WC leverages multi-view inputs, it still suffers from frequent knock-offs due to insufficient local perception. The FTC policy is weaker during the approach phase because it lacks a true global perception (e.g., a static third-view). However, once it reaches the box, it tracks the stick center accurately, establishes stable contact, and completes the task. Adding a wrist view (FTC w/ WC) does not resolve the localization issue and yields performance comparable to FTC. HT also fails in some trials due to unstable hand tracking and network instability.

3) *Curtain-Occluded Object Retrieval:* In this task, the robot is required to retrieve an object from a cabinet partially occluded by a hanging curtain, as shown in Fig. 4 (c). The robot should establish safe interaction with the deformable curtain to create sufficient clearance for actions. Task success is defined as removing the object from the cabinet without it falling down.

Challenges: Local perception: The object is placed inside the cabinet behind the hanging curtain, making local perception critical for guiding the hand to localize and grasp the target object. Visual occlusions: Due to the hanging curtain, the object can only be observed through an opening at the bottom of the cabinet side. The top-down third-view camera can hardly observe the target object, and interaction with the curtain will block the other camera views during the robot execution phase.

Test scenarios: We conduct **50** rollouts for each policy. As

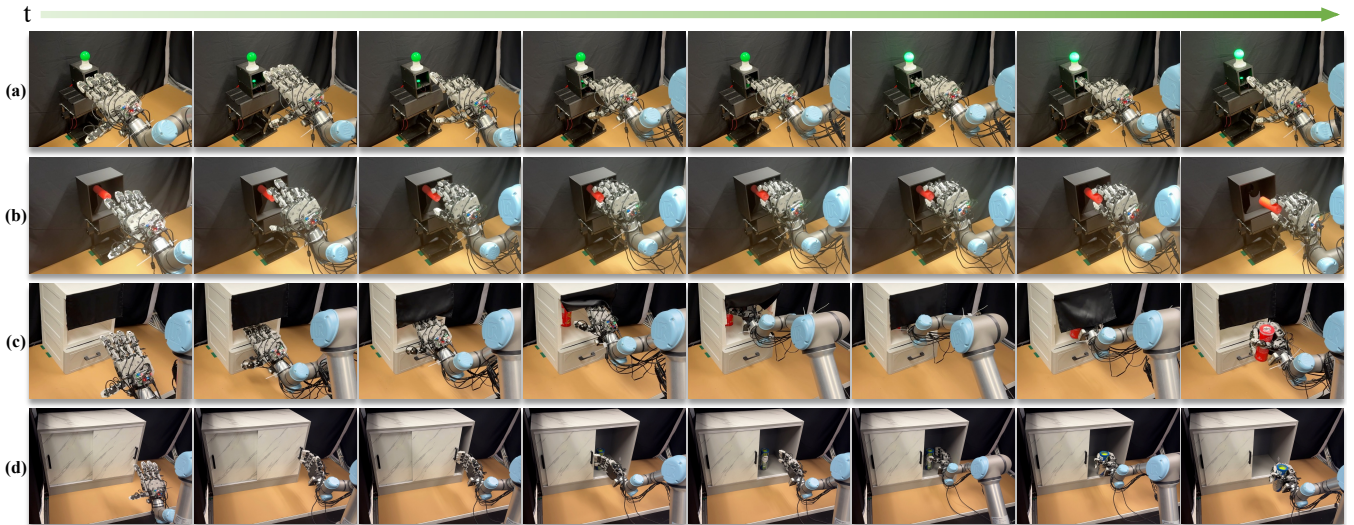


Fig. 5: **Examples of the Four Dexterous Manipulation Tasks.** The pictures on the far left and right show the initial and final states, respectively. Each row shows the task progression over time.

TABLE I: **Comparison with Baselines Across Diverse Real-World Tasks.**

Methods	Confined-Box Button Pressing	Unstable-Support Stick Retrieval	Curtain-Occluded Object Retrieval	Closed-Cabinet Object Retrieval		Average Success Rate
				Open Door	Retrieve Object	
TVC	8/42 (19.0%)	14/37 (37.8%)	20/50 (40.0%)	44/55 (80.0%)	30/55 (54.5%)	37.8%
WC	2/42 (4.8%)	10/37 (27.0%)	28/50 (56.0%)	47/55 (85.5%)	34/55 (61.8%)	37.4%
TVC w/ WC	9/42 (21.4%)	17/37 (45.9%)	34/50 (68.0%)	49/55 (89.1%)	39/55 (70.9%)	51.6%
FTC	24/42 (57.1%)	21/37 (56.8%)	37/50 (74.0%)	48/55 (87.3%)	37/55 (67.3%)	63.8%
FTC w/ WC	18/42 (42.9%)	19/37 (51.4%)	38/50 (76.0%)	51/55 (92.7%)	40/55 (72.7%)	60.8%
HT	42/42 (100.0%)	34/37 (91.9%)	48/50 (96.0%)	55/55 (100.0%)	54/55 (98.2%)	96.5%
w/o camera poses	22/42 (52.4%)	20/37 (54.1%)	39/50 (78.0%)	44/55 (80.0%)	37/55 (67.3%)	63.0%
w/o joint currents	24/42 (57.1%)	22/37 (59.5%)	42/50 (84.0%)	47/55 (85.5%)	41/55 (74.5%)	68.8%
Ours	31/42 (73.8%)	28/37 (75.7%)	45/50 (90.0%)	50/55 (90.9%)	46/55 (83.6%)	80.8%

shown in Fig. 4 (c), the test configurations can be grouped into six categories: T1) Variations in initial object poses (20 rollouts); T2) Unseen target objects (10 rollouts); T3) Curtains with unseen materials (5 rollouts); T4) Curtain in unseen color (5 rollouts); T5) Curtains with unseen lengths (5 rollouts); and T6) Unseen cabinet wall appearance (stickers; 5 rollouts).

Performance: The training dataset contains 245 demonstration episodes, with each demonstration averaging 8.5s. We report both qualitative (Fig. 5 (c)) and quantitative results (Table I), our method achieves an overall 90.0% success rate, outperforming all baselines.

In this task, occlusion from the hanging curtain limits the effectiveness of the *TVC* approach. When the object is placed deep inside or close to the cabinet’s left side and becomes fully hidden by the curtain and the cabinet wall, the hand is unable to reach the correct position, causing grasp failures. A single wrist-mounted camera (*WC*) is also insufficient because the thumb can occlude the view during grasping, leading to unstable grasps. And without reliable global scene awareness, collisions with the cabinet can trigger the arm’s e-stop. Using both third-view and wrist-mounted cameras (*TVC w/ WC*) reduces e-stop events, but limited local perception remains a bottleneck. The *FTC* and *FTC w/ WC* policies localize the object reliably, yet they still often collide with the cabinet bottom during retrieval,

triggering e-stops and terminating execution.

4) *Closed-Cabinet Object Retrieval:* In this long-horizon task, the robot is required to retrieve a target object from a closed cabinet, as shown in Fig. 4 (d). The robot needs to reach the door handle, open the cabinet, grasp the target object, and safely retrieve it. In this task, we define two success metrics: successfully opening the cabinet and successfully retrieving the object without collision.

Challenges: Long-horizon manipulation: The robot first extends its hand and rotates the wrist to orient the fingertips toward the handle. It then gently makes contact and slides the cabinet door, followed by pushing the door open to create sufficient clearance for safe entry. Finally, the robot reaches into the cabinet, grasps the target object, and retrieves it. Local perceptual capability: Given the handle’s small size, the robot needs fine-grained, close-range perception to accurately localize the handle and establish contact. Precision and contact awareness: Door opening requires precise position control together with contact-aware control to maintain stable contact with the door. The robot must regulate the pushing distance and interaction force to generate enough clearance while avoiding damage to the cabinet. Visual occlusions: The target object inside the cabinet is often heavily occluded and may be difficult to be observed

from a fixed top-down third-view camera.

Test scenarios: We conduct **55** rollouts for each policy. As shown in Fig. 4 (d), the test configurations can be grouped into five categories: *T1*) Variations in object initial poses, seen during training (25 rollouts); *T2*) Unseen target objects (15 rollouts); *T3*) Cabinet handle with unseen shapes (5 rollouts); *T4*) Cabinet handle in unseen colors (5 rollouts); and *T5*) Unseen cabinet wall appearance (stickers; 5 rollouts).

Performance: The training dataset contains 256 demonstration episodes, with each demonstration averaging 13.5 s. We report both qualitative (Fig. 5 (d)) and quantitative results (Table I), our method achieves an overall 83.6% success rate, outperforming all baselines.

During door opening, the *TVC* policy may extend the fingers behind the door near the handle and get jammed, leading to opening failures. The object is often only partially visible and can become fully occluded with pose variations, resulting in a low retrieval success rate. The *WC* policy supports door opening, but finger–handle misalignment can still occur. During in-cabinet grasping, the thumb may occlude the wrist camera, and without global context the door is sometimes not opened wide enough. During retrieval, the object may collide with the cabinet bottom or the door, causing failures or e-stops. Combining third-view and wrist views (*TVC w/ WC*) helps open the door to the desired position, but grasping the object inside the cabinet remains unstable due to thumb occlusion of the wrist camera and object-induced occlusion of the wrist view. Fingertip-based policies (*FTC* and *FTC w/ WC*) perform well in door opening, but inadequate opening and collision-induced e-stops can still occur. This is consistent with limited global awareness of door opening state and bottom clearance during retrieval.

C. Ablation Studies

To illustrate the importance of camera pose encoding and finger joint current encoding modules of our learning system, we conduct two ablation experiments across the four tasks, with the same training dataset and evaluation scenarios: (i) *w/o camera poses*, where the policy is trained without camera pose inputs; (ii) *w/o joint currents*, where the policy is trained without finger joint current inputs.

As shown in Table I, our results reveal the critical importance of camera poses and finger current signals for learning a dexterous manipulation policy. Specifically, for contact-rich tasks such as button pressing and object grasping, removing finger current inputs significantly reduces success rates, highlighting the essential role of contact in understanding physical interactions between the robotic hand and objects. On the other hand, for tasks that require continuous and dexterous motion of the hand, such as unstable-support stick retrieval, incorporating camera pose encoding for dynamically moving fingertip cameras notably enhances multi-view perception, which improves the policy’s ability to execute accurate movement to center and conform the stick. These findings emphasize the necessity of both finger currents and camera poses to achieve robust dexterous manipulation.

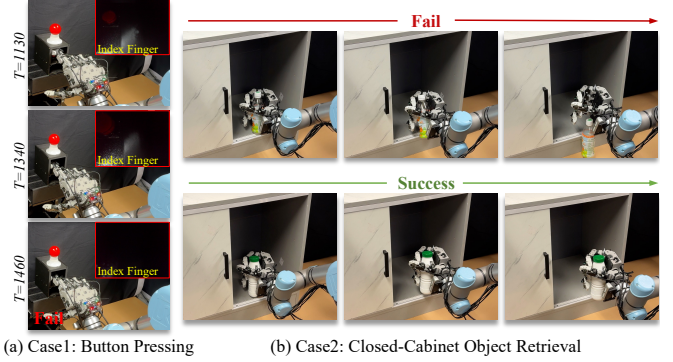


Fig. 6: **Failure Cases.** (a) Case 1 shows a failure in the non-illuminated-button setting of the confined-box button-pressing task; (b) Case 2 includes two examples in the closed-cabinet object retrieval task: one failure with an unseen, slippery object and one success with a rough object seen during training.

V. DISCUSSION AND FUTURE WORK

Discussion. We analyze two representative failure cases of FingerViP (Fig. 6). *Case 1:* Confined-box button pressing. When pressing a non-illuminated button, the fingertip view becomes nearly dark after insertion, making the button hard to perceive and localize, leading to repeated misalignment and task failure. A potential solution is to integrate a small light source at the fingertip and apply low-light data augmentation during training. *Case 2:* Closed-cabinet object retrieval. We observed that FingerViP often fails on a slender, slippery juice bottle that tends to slip out of the hand during grasping and retrieval. Nevertheless, FingerViP performs well on a similar sized white bottle with a rough, matte surface. We hypothesize that the lack of high-friction patches on the hand makes it easier for slender and slippery objects to slip out under grasping forces. Adding high-friction patches to the hand may help mitigate this failure case.

Future work. First, we will further compact the fingertip module and integrate a small fingertip light source for low-light operation. Second, we will leverage the calibrated multi-view fingertip setup to enable stable local 3D reconstruction with known camera poses, providing geometry-aware state estimates for policy learning. Third, we also plan to integrate compact tactile sensing to capture complementary fine-grained contact signals for more precise manipulation.

VI. CONCLUSIONS

In this paper, we present FingerViP, a learning system for real-world dexterous manipulation with fingertip visual perception. First, we design a vision-enhanced fingertip module with an embedded miniature camera and mount the modules on a multi-fingered hand to provide comprehensive, multi-view observations of both the hand and the surrounding environment, mitigating occlusions. Second, we develop a visuomotor policy that leverages third-view and multi-view fingertip observations augmented with fingertip-camera pose and per-finger joint-current encodings to improve view–proprioception alignment and contact awareness, enabling robust dexterous manipulation.

FingerViP significantly outperforms baselines on four challenging real-world dexterous manipulation tasks, demonstrating high efficiency and strong generalization, and highlighting the importance of multi-view fingertip visual perception. Overall, our system offers a scalable multi-view perception approach for multi-fingered dexterous manipulation. To support further research and contribute to the community, we will open-source all hardware designs and code.

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